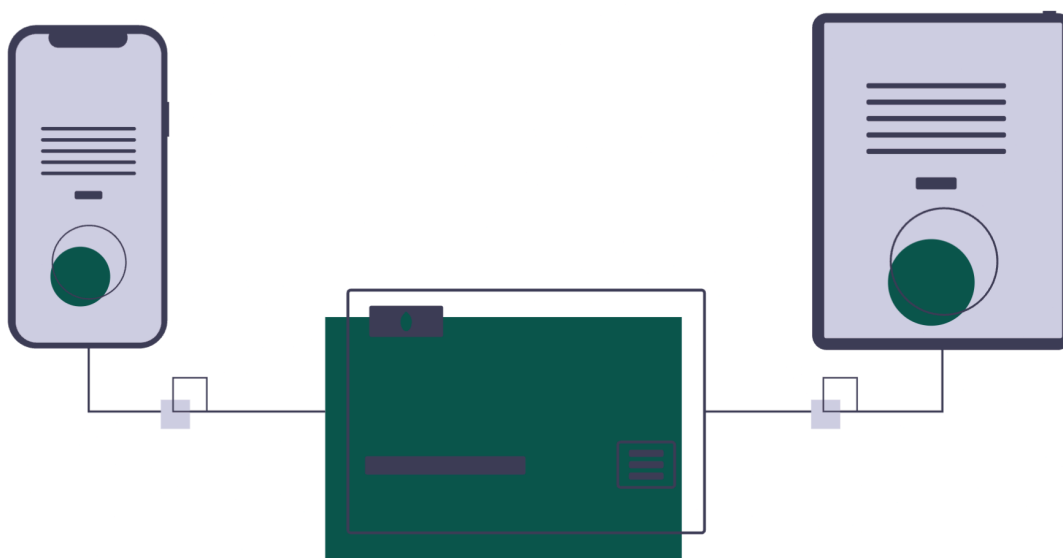


Mini Project of CSE405

Computer Networks



Submitted By:

Aklhak Hossain

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<https://akhlak.com>

Submitted To:

Md. Khalid Mahbub Khan

Senior Lecturer

Department of CSE

East West University

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1. Problem Statement:

Modern universities operate across multiple campuses and depend on a robust, interconnected network to support academic and administrative functions. Avalon University, a large educational institution similar to East West University, hosts a diverse range of network-dependent services, including online admissions, academic advising, results management, e-Tendering, library services, and financial accounts, all of which require reliable, low-latency, and scalable connectivity.

The challenge addressed in this project is designing a complete multi-campus university network from scratch. The network must interconnect four physically separate campuses (each with its own Local Area Network) so that every host across all campuses can communicate seamlessly. The design must adhere to a structured IP addressing scheme using Class A addresses for LAN segments and Class C addresses for WAN inter-router links, with subnetting applied as necessary. Static routing is used to manually define the path traffic takes through the network, giving full administrative control over routing decisions without the overhead of dynamic routing protocols.

The network was designed and simulated using Cisco Packet Tracer.

2. Design & Experimental Results

2.1 Network Topology

The network consists of four campuses, each represented by one router, one switch, and three host PCs. The four routers are connected in a linear chain using serial WAN links, forming a path:

Router 0: Connected to R1 and R3. Hosts the Web Server.

Router 1: Connected to R0 and R2. Hosts the DNS Server.

Router 2: Connected to R1 and R3. Host the DHCP Server.

Router 3: Connected to R2 and R0

Devices used:

- 4 × Router-PT
- 4 × Switch-PT
- 9 × PCs
- 3 × SERVER-PT
- 4 × Serial DCE/DTE cable pairs (WAN links)
- 4 × Copper Straight-Through cables (router to switch)
- 12 × Copper Straight-Through cables (DEVICESs to switches)

2.2 DIAGRAM

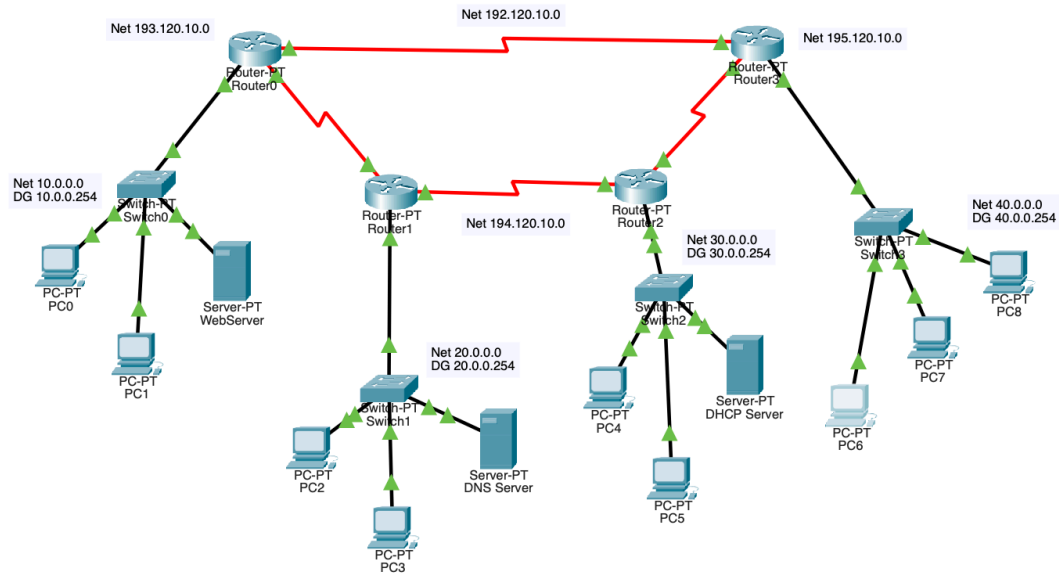


Figure: Network Topology

2.3 IP Addressing Scheme:

Router	Interface	IP Address	Subnet Mask	Description
Router 0	fa0/0	10.0.0.254	255.0.0.0	LAN 1
	se2/0	193.120.10.1	255.255.255.0	Link to Router 1
	se3/0	192.120.10.2	255.255.255.0	Link to Router 3
Router 1	fa0/0	20.0.0.254	255.0.0.0	LAN 2
	se2/0	194.120.10.1	255.255.255.0	Link to Router 0
	se3/0	193.120.10.2	255.255.255.0	Link to Router 2
Router 2	fa0/0	30.0.0.254	255.0.0.0	LAN 3
	se2/0	195.120.10.1	255.255.255.0	Link to Router 1
	se3/0	194.120.10.2	255.255.255.0	Link to Router 3
Router 3	fa0/0	40.0.0.254	255.0.0.0	LAN 4
	se2/0	192.120.10.1	255.255.255.0	Link to Router 2
	se3/0	195.120.10.2	255.255.255.0	Link to Router 0

3. Experimental Results:

3.1 Routing Configuration:

The document presents the complete Command Line Interface (CLI) configuration details for every router in the network topology. Static routing was configured on all four routers to ensure full visibility across the network.

Router 0:

```
enable
config
interface fa0/0
ip address 10.0.0.254 255.0.0.0
no shut
do wr
exit
interface se2/0
ip address 193.120.10.1 255.255.255.0
clock rate 64000
no shut
do wr
exit
interface se3/0
ip address 192.120.10.2 255.255.255.0
no shut
do wr
exit
```

OSPF 1:

```
router ospf 1
network 10.0.0.0 0.255.255.255 area 1
network 192.120.10.0 0.0.0.255 area 1
network 193.120.10.0 0.0.0.255 area 1
exit
```

Router 1:

```
enable
config
interface fa0/0
ip address 20.0.0.254 255.0.0.0
```

```
no shut
do wr
exit
interface se2/0
ip address 194.120.10.1 255.255.255.0
clock rate 64000
no shut
do wr
exit
interface se3/0
ip address 193.120.10.2 255.255.255.0
no shut
do wr
exit
```

OSPF 2:

```
router ospf 1
network 20.0.0.0 0.255.255.255 area 1
network 193.120.10.0 0.0.0.255 area 1
network 194.120.10.0 0.0.0.255 area 1
exit
```

Router 2:

```
enable
config
interface fa0/0
ip address 30.0.0.254 255.0.0.0
no shut
do wr
exit
interface se2/0
ip address 195.120.10.1 255.255.255.0
clock rate 64000
no shut
do wr
exit
interface se3/0
ip address 194.120.10.2 255.255.255.0
no shut
do wr
exit
```

OSPF 3:

```
router ospf 1
network 30.0.0.0 0.255.255.255 area 1
network 194.120.10.0 0.0.0.255 area 1
network 195.120.10.0 0.0.0.255 area 1
exit
```

Router 3:

```
enable
config
interface fa0/0
ip address 40.0.0.254 255.0.0.0
no shut
do wr
exit
interface se2/0
ip address 192.120.10.1 255.255.255.0
clock rate 64000
no shut
do wr
exit
interface se3/0
ip address 195.120.10.2 255.255.255.0
no shut
do wr
exit
```

OSPF 4:

```
router ospf 1
network 40.0.0.0 0.255.255.255 area 1
network 195.120.10.0 0.0.0.255 area 1
network 192.120.10.0 0.0.0.255 area 1
exit
```

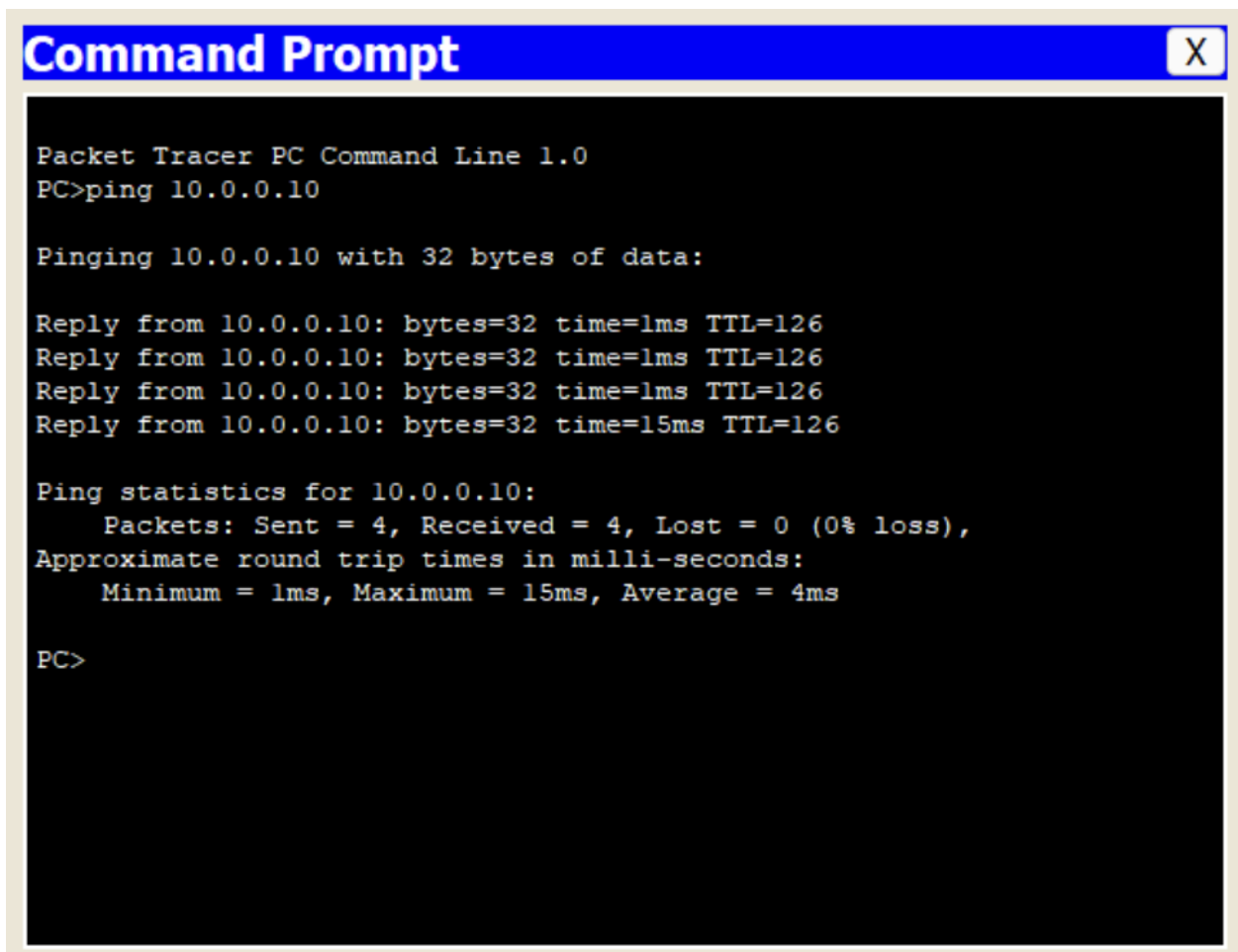
3.2 Server Configuration Results:

- **Web Server:** Configured on 10.0.0.10. The HTTP service is active, serving a page with the student name "Welcome to Avalon University " as required.
- **DNS Server:** Configured on 20.0.0.10. It successfully maps www.avalon. edu to the Web Server IP 10.0.0.10.
- **DHCP Servers:** Independent DHCP pools were created for each campus. PCs in every campus successfully requested and received dynamic IP addresses from their local DHCP server.

3.3 Server Configuration Results:

Ping tests were conducted from Campus A hosts to verify inter-campus connectivity. All tests returned successful replies with 0% packet loss.

Test 1: Campus A to Campus B:



```
Command Prompt X

Packet Tracer PC Command Line 1.0
PC>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=1ms TTL=126
Reply from 10.0.0.10: bytes=32 time=1ms TTL=126
Reply from 10.0.0.10: bytes=32 time=1ms TTL=126
Reply from 10.0.0.10: bytes=32 time=15ms TTL=126

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 15ms, Average = 4ms

PC>
```


Test 2: Campus A to Campus B:



4. Conclusion:

This project successfully demonstrated the design and simulation of a multi-campus university network for Avalon University using Cisco Packet Tracer. A total of four LANs were created — one per campus, each running on a Class A subnet (x.0.0.0) with three wired host PCs. The four campuses were interconnected through a chain of four routers linked by Class C /30 WAN serial links (x.120.x.10), providing efficient use of IP address space on point-to-point connections.

Static routing was used to establish full reachability across all campuses. Each router was manually configured with routes to every remote network, directing traffic to the appropriate next-hop IP. This approach gives the network administrator complete control over traffic paths and eliminates the processing overhead associated with dynamic routing protocols such as OSPF, which is appropriate for a small, stable topology like this one.

Connectivity was verified through successful ping operations between all four campuses, and PDU simulation confirmed that packets follow the correct path through the network. The results validate that the design meets all stated requirements: four LANs, full inter-campus connectivity, Class A and C addressing, subnetting, and static routing.

In a real-world deployment, this static design could be extended with dynamic routing, VLANs for departmental segmentation, firewall policies between campuses, and redundant WAN links for fault tolerance.